

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the Human Development Index (HDI) Prediction Using Machine Learning project. Each feature contributes to the overall functionality and usability of the application.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Dataset Loading	Loads the HDI dataset from a CSV file using	Data Processing	Done	High
2	Data Preprocessing	Handles missing values and prepares the	Data Processing	Done	High
3	Machine Learning Model	Trains a Linear Regression model for HDI	ML Model	Done	High
4	HDI Prediction	Predicts the HDI score based on user inputs	Prediction Engine	Done	High
5	Flask Web Application	Connects the trained model with a web	Backend	Done	High
6	User Interface	Accepts user input and displays	Frontend (HTML/CSS)	Done	Medium
7	Model Saving & Loading	Saves the trained model as model.pkl and	Model Management	Done	Medium
8	Result Display	Displays predicted HDI score and	Output Module	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form, Result display
2	Backend / Logic	3	Flask backend, Prediction logic, Model loading
3	Database / Storage	1	CSV dataset and model.pkl storage
4	API / Integration	1	Flask integration with ML model
5	Security / Authentication	1	Input validation and error handling